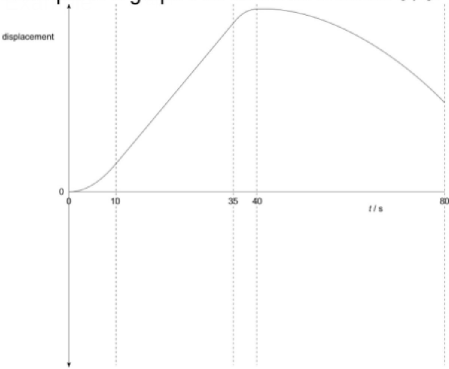


| Question  | Answer                                                                                                                                        | Marks       | Guidance                                                                                                                                                                                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1(a)      | amplitude and power identified as scalars                                                                                                     | <b>B1</b>   | Allow methods for marking the table other than a tick (e.g. ×) provided no ticks are seen                                                                                                                                                                                                                                            |
|           | force and momentum identified as vectors                                                                                                      | <b>B1</b>   |                                                                                                                                                                                                                                                                                                                                      |
| 1(b)(i)   | area (under graph line)                                                                                                                       | <b>B1</b>   |                                                                                                                                                                                                                                                                                                                                      |
| 1(b)(ii)  | distance = 390 m                                                                                                                              | <b>A1</b>   |                                                                                                                                                                                                                                                                                                                                      |
| 1(b)(iii) | $a = (v - u) / t$ or $a = \Delta v / (\Delta t)$ or gradient (of graph)<br>e.g. = $(0 - 12) / 5$                                              | <b>C1</b>   | Allow $a = (u - v) / t$ or $a = (\Delta v) / (\Delta t)$ or $a = -\text{gradient}$<br>Allow $a = (y_2 - y_1) / (x_2 - x_1)$<br>Allow a numerical expression equivalent to one of the three non-zero gradients on the graph, e.g. $a = 12 / 5$ or $a = 12 / 10$ or $a = 10 / 40$<br>No ECF of incorrect velocity from <b>1(b)(ii)</b> |
|           | magnitude of acceleration = $2.4 \text{ m s}^{-2}$                                                                                            | <b>A1</b>   | Allow $-2.4 \text{ m s}^{-2}$                                                                                                                                                                                                                                                                                                        |
| 1(b)(iv)  | $v^2 = 2 \times -0.25 \times -150$<br>$= 75$                                                                                                  | <b>C1</b>   | No mark for symbol formulae. For each method, the C1 mark is given for the full substitution of data into the formula.<br>Allow $v^2 = 2 \times 0.25 \times 150$                                                                                                                                                                     |
|           | speed = $8.7 \text{ m s}^{-1}$                                                                                                                | <b>A1</b>   | Allow speed = $-8.7 \text{ m s}^{-1}$ for all methods                                                                                                                                                                                                                                                                                |
|           | <u>Alternative methods:</u><br>$(-150 = \frac{1}{2} \times -0.25 \times t^2 \text{ so } t = 34.6)$<br><br>$v = -0.25 \times 34.6$<br>$= -8.7$ | <b>(C1)</b> | Allow $v = 0.25 \times 34.6$<br>Allow use of $t = 35$ to imply the C1 mark, but do not credit the A1 mark if leads to incorrect speed ( $8.8 \text{ m s}^{-1}$ )                                                                                                                                                                     |
|           | speed = $8.7 \text{ m s}^{-1}$                                                                                                                | <b>(A1)</b> |                                                                                                                                                                                                                                                                                                                                      |

| Question | Answer                                                                                  | Marks | Guidance                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1(b)(iv) | OR<br>$-150 = \frac{1}{2} \times v \times 34.6$<br>$v = -8.7$                           | (C1)  | Allow $150 = \frac{1}{2} \times v \times 34.6$<br>Allow use of $t = 35$ to imply the C1 mark, but do not credit the A1 mark if leads to incorrect speed ( $8.6 \text{ m s}^{-1}$ ) |
|          | speed = $8.7 \text{ m s}^{-1}$                                                          | (A1)  |                                                                                                                                                                                    |
|          | OR<br>$-150 = (v \times 34.6) - (\frac{1}{2} \times -0.25 \times 34.6^2)$<br>$v = -8.7$ | (C1)  | Allow $150 = (v \times 34.6) - (\frac{1}{2} \times 0.25 \times 34.6^2)$                                                                                                            |
|          | speed = $8.7 \text{ m s}^{-1}$                                                          | (A1)  |                                                                                                                                                                                    |
|          | OR<br>$t = 34.6 + 40$<br>$= 74.6$<br>At $t = 74.6$ , $v = -8.7$ (from graph)            | (C1)  | No tolerance (for consistency with other methods)                                                                                                                                  |
|          | speed = $8.7 \text{ m s}^{-1}$                                                          | (A1)  |                                                                                                                                                                                    |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                          | Marks     | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1(b)(v)  | <p>A line from 0,0 to <math>t = 80</math> s with increasing displacement then decreasing displacement that does not cross the <math>t</math>-axis</p>                                                                                                                                                                                                                                                           | <b>B1</b> | <p>For all marking points ignore line after <math>t = 80</math> s</p> <p>Allow line to have discontinuous gradient at <math>t = 10, 35, 40</math></p> <p>Do not allow step change in displacement</p> <p>Allow graph to end up to 5 mm before <math>t = 80</math> s</p> <p>Allow line to end at the <math>t</math>-axis</p> <p>Do not allow displacement that increases, decreases and then increases again</p>             |
|          | A line with a maximum displacement at $t = 40$ s                                                                                                                                                                                                                                                                                                                                                                | <b>B1</b> | Allow centre of peak $\pm 3$ mm horizontally from $t = 40$ s                                                                                                                                                                                                                                                                                                                                                                |
|          | <p>A line with</p> <ul style="list-style-type: none"> <li>increasing gradient from <math>t = 0</math> to <math>t = 10</math> s</li> <li>constant positive gradient between <math>t = 10</math> s and <math>t = 35</math> s</li> <li>decreasing gradient between <math>t = 35</math> s and <math>t = 40</math> s</li> <li>decreasing gradient between <math>t = 40</math> s and <math>t = 80</math> s</li> </ul> | <b>B1</b> | <p>Allow line to have discontinuous gradient at <math>t = 10, 35, 40</math></p> <p>Do not allow step change in displacement</p> <p>If unclear whether a section of the line is straight or curved give BOD</p> <p>Allow graph to end up to 5 mm before <math>t = 80</math> s</p> <p>Example of a graph that would be credited 3 / 3:</p>  |